Latest Fig.10 VDF moments compared with Fig.11

(a) Experiment

(b) Simulation without probe

(c) Simulation with probe

Electron temperature (eV)

Fig.11 experimental markers
Inferred e: full signed moment

Fig.11 simulation ink envelope
Inferred e: full signed moment

Fig.11 simulation ink envelope
Inferred e: full signed moment

Ion temperature (eV)

Fig.11 experimental markers
Inferred H: full signed moment
Inferred C: full signed moment

Fig.11 simulation ink envelope
Inferred H: full signed moment
Inferred C: full signed moment

Fig.11 simulation ink envelope
Inferred H: full signed moment
Inferred C: full signed moment

Time (ns)

Original inferred VDFs; 4097 velocity samples per species; no clipping, smoothing or temperature fitting.
Fig.11 gives one ion temperature, not separate H/C truth. Shading reflects spatial inhomogeneity; digitized ink is approximate.

Projected flow velocity from the latest inferred VDFs

(a) Experiment

(b) Simulation without probe

(c) Simulation with probe

Electrons flow (km/s)

Hydrogen flow (km/s)

Carbon flow (km/s)

Time (ns)

Signed first moments on the saved velocity axis. Fig.11 has no flow data; dashed zero is a visual reference, not a measured truth.

Spectral closure on the latest native wavelength grids
Existing OTS_spectrum_PV; fixed $n_e = 3.5 \times 10^{20}$ cm^{-3}; authorized VDF copies

Latest data / row peak

Existing PV reconstruction

PV minus latest data

(a) Experiment
Time (ns)

(b) Simulation without probe
Time (ns)

(c) Simulation with probe
Time (ns)

Mean relative L2: 55.9%

Mean relative L2: 36.5%

Mean relative L2: 31.3%

Normalized intensity

Normalized intensity difference

$\Delta\lambda = \lambda - 263.3$ nm

$\Delta\lambda = \lambda - 263.3$ nm

$\Delta\lambda = \lambda - 263.3$ nm

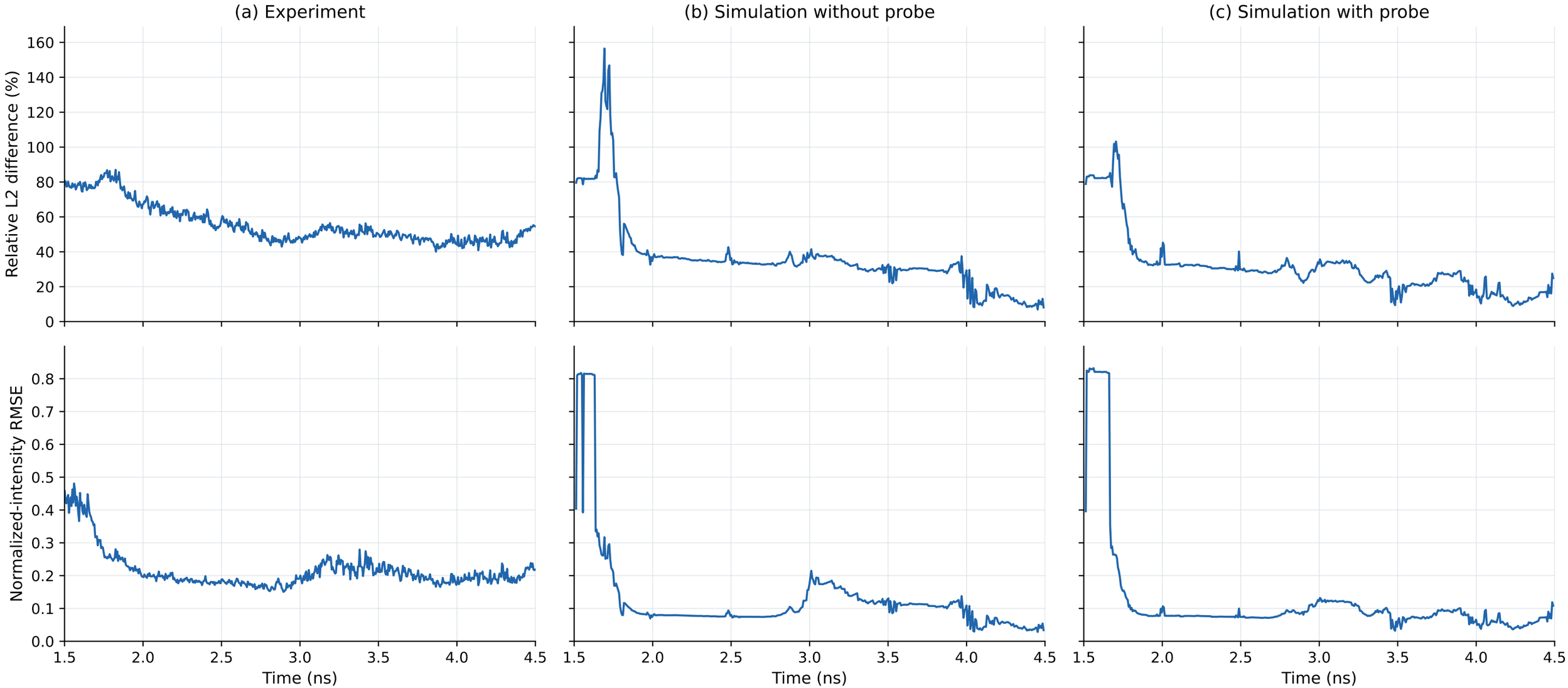
Representative spectra nearest 2, 3 and 4 ns

Latest data / row peak
Existing PV reconstruction

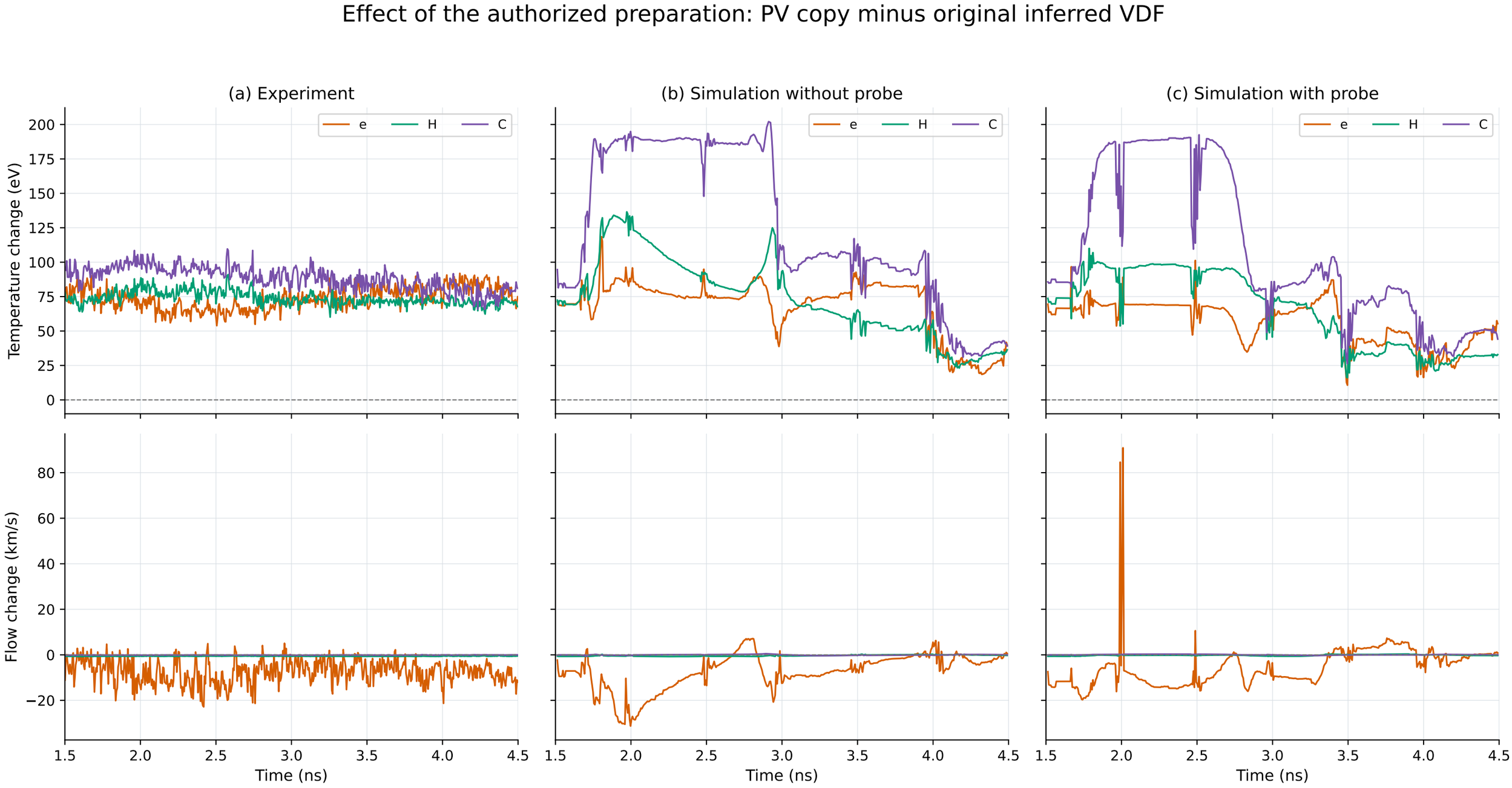
1.998000 ns | source row 236
Rel. L2 = 67.3%
3.002400 ns | source row 422
Rel. L2 = 47.8%
4.001400 ns | source row 607
Rel. L2 = 43.9%
1.997409 ns | source row 93
Rel. L2 = 34.7%
2.998446 ns | source row 254
Rel. L2 = 37.2%
3.999482 ns | source row 415
Rel. L2 = 29.4%
1.997409 ns | source row 93
Rel. L2 = 37.1%
2.998446 ns | source row 254
Rel. L2 = 33.1%
3.999482 ns | source row 415
Rel. L2 = 18.4%

(a) Experiment
Normalized intensity
(b) Simulation without probe
Normalized intensity
(c) Simulation with probe
Normalized intensity
0.0
0.2
0.4
0.6
0.8
1.0
−1.0
−0.5
0.0
0.5
1.0
$\Delta\lambda = \lambda - 263.3$ nm

Native wavelength axes; independent row-peak normalization. Fixed density; no spectral shift, smoothing, fit or instrument convolution.



Relative L2 = ||PV - normalized data||₂ / ||normalized data||₂. Comparison of spectral shape, not absolute optical power.


Effect of the authorized preparation: PV copy minus original inferred VDF
(a) Experiment
(b) Simulation without probe
(c) Simulation with probe
e
H
C
Temperature change (eV)
Flow change (km/s)
Time (ns)
Uniform float64 grid → linear interpolation → negative values set to zero → unit integral. Original inference arrays remain unchanged.